

Task 2: Descriptive Statistics and Correlation

Using the traffic dataset, calculate:

2.1 Traffic volume statistics

- Mean
- Median
- Standard deviation
- Variance
- Range

Interpret what these statistics tell you about traffic volume and its variability.

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```
1 SELECT
2   ROUND(AVG(traffic_volume), 2) AS mean,
3   MIN(traffic_volume) AS minimum,
4   MAX(traffic_volume) AS maximum,
5   MAX(traffic_volume) - MIN(traffic_volume) AS range
6 FROM traffic;
```

1 row

mean	minimum	maximum	range
3259.82	0	7280	7280

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```
1 SELECT AVG(traffic_volume) AS median
2 FROM (
3   SELECT traffic_volume
4   FROM traffic
5   ORDER BY traffic_volume
6   LIMIT 2
7   OFFSET 24101
8 );
```

1 row

median
3380

```
1 SELECT
2   ROUND(AVG(traffic_volume * traffic_volume)
3     - AVG(traffic_volume) * AVG(traffic_volume), 2) AS variance
4 FROM traffic;
```

1 row

variance
3947533.43

Standard deviation is:

$$SD = \sqrt{Variance}$$

Therefore:

$$SD = \sqrt{3,947,533.43} = 1,986.84$$

Task 2.1 — Final Results

Statistic	Result 
Mean	3,259.82 vehicles
Median	3,380 vehicles
Standard deviation	1,986.84 vehicles
Variance	3,947,533.43 vehicles ²
Range	7,280 vehicles

The mean traffic volume was 3,259.82 vehicles per hour, while the median was slightly higher at 3,380 vehicles, indicating that the typical hourly traffic volume was around 3,300 vehicles. The standard deviation of 1,986.84 vehicles is relatively large compared with the mean, showing substantial variation in traffic volume across different observations. Traffic volume ranged from 0 to 7,280 vehicles, giving a range of 7,280 vehicles. Overall, the results indicate that traffic demand varies considerably, which is consistent with different traffic conditions occurring across hours, days and periods of the year.

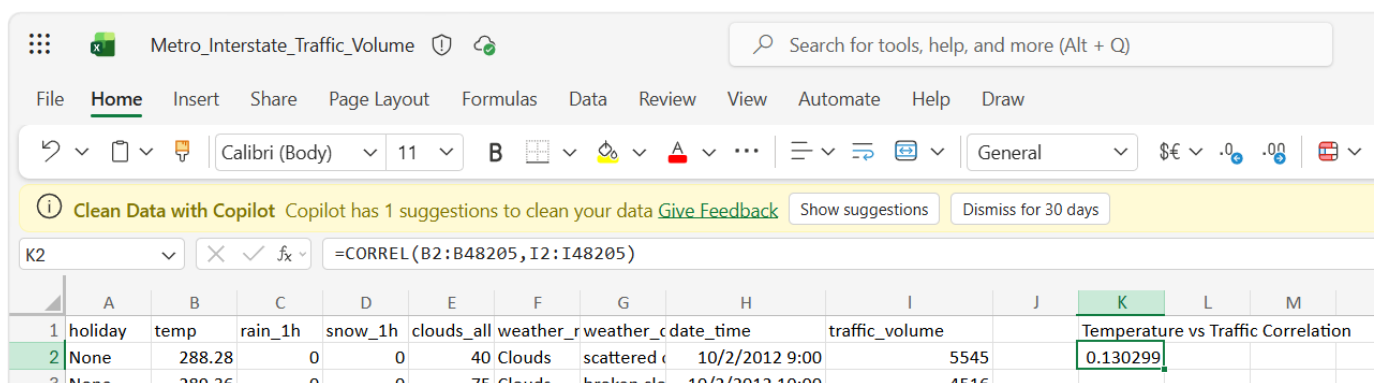
2.2 Correlation analysis

Calculate the correlation coefficient between:

- Temperature
- Traffic volume

Then:

- Interpret the **direction** of the relationship.
- Interpret the **strength** of the relationship.
- Explain why correlation does not necessarily imply causation.



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	holiday	temp	rain_1h	snow_1h	clouds_all	weather_r	weather_c	date_time	traffic_volume				
2	None	288.28	0	0	40	Clouds	scattered c	10/2/2012 9:00	5545		0.130299		
3	None	289.36	0	0	75	Clouds	broken cl	10/2/2012 10:00	4516				

Task 2.2 — Final Answer

Item	Result / Interpretation
Variables	Temperature and Traffic Volume
Method	Pearson correlation using Excel <code>CORREL</code>
Correlation coefficient	$r = 0.130299 \approx 0.1303$
Direction	Positive
Strength	Weak

Direction

The correlation coefficient is positive ($r = 0.1303$), indicating a **positive relationship** between temperature and traffic volume. In general, traffic volume tends to increase slightly as temperature increases.

Strength

The relationship is **weak** because the correlation coefficient is close to zero. This means temperature has only a small linear association with traffic volume and, by itself, isn't a strong indicator of traffic demand.

Why correlation doesn't imply causation

Although temperature and traffic volume have a weak positive correlation, this **doesn't mean that higher temperatures cause traffic volume to increase**. Traffic volume may be influenced by many other factors, including time of day, day of the week, holidays, weather conditions and seasonal travel patterns. Correlation measures the association between two variables but doesn't establish a cause-and-effect relationship.

Task 3: Probability and Congestion Analysis

Define **congestion** as: traffic volume $> 5,500$ vehicles.

Calculate:

3.1 Basic probability

- **P(Congestion)**
- **P(Clear Weather)**

- **P(Congestion AND Clear Weather)**

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```

1 SELECT
2   COUNT(*) AS congestion_records
3 FROM traffic
4 WHERE traffic_volume > 5500;

```

1 row

Database file▼
Queries▼
Save & Share
Export▼
Run ▶

```

1 SELECT
2   COUNT(*) AS clear_records
3 FROM traffic
4 WHERE weather_main = 'Clear';

```

1 row

congestion_records
7100

clear_records
13391

```

1 SELECT
2   COUNT(*) AS congestion_and_clear
3 FROM traffic
4 WHERE traffic_volume > 5500
5 AND weather_main = 'Clear';

```

1 row

congestion_and_clear
1763

Event	Records	Probability
Congestion (traffic_volume > 5500)	7,100	0.1473 = 14.73%
Clear Weather	13,391	0.2778 = 27.78%
Congestion AND Clear Weather	1,763	0.0366 = 3.66%

Probability	Calculation
P(Congestion)	Congestion records ÷ 48,204
P(Clear Weather)	Clear records ÷ 48,204
P(Congestion AND Clear)	Congestion & Clear records ÷ 48,204

Approximately 14.73% of the observations meet the congestion definition. Clear weather occurs in 27.78% of observations. The probability of experiencing both congestion and clear weather is 3.66%.

3.2 Conditional probability

Calculate:

- **P(Clear Weather | Congestion)**
- **P(High Temperature | Congestion)**, where high temperature > 292K
- Determine whether weather conditions and congestion can be considered independent using $P(A \cap B) = P(A) \times P(B)$
- Calculate the **odds ratio of congestion in clear versus cloudy weather**.

Conclude with a short explanation of what the probability analysis suggests about the relationship between weather and congestion.

- **P(Clear Weather | Congestion) / P(High Temperature | Congestion)**, where high temperature > 292K

<pre>1 SELECT 2 1763.0 / 7100 AS probability;</pre>	<pre>1 SELECT 2 COUNT(*) AS high_temp_congestion 3 FROM traffic 4 WHERE traffic_volume > 5500 5 AND temp > 292;</pre>				
1 row	1 row				
<table><tr><th>probability</th></tr><tr><td>0.2483098591549296</td></tr></table>	probability	0.2483098591549296	<table><tr><th>high_temp_congestion</th></tr><tr><td>1867</td></tr></table>	high_temp_congestion	1867
probability					
0.2483098591549296					
high_temp_congestion					
1867					

Among observations classified as congested, approximately 24.83% occurred during clear weather.

Approximately 26.30% of congested observations occurred when the temperature was above 292 K.

<pre>1 SELECT 2 (7100.0 / 48204) * (13391.0 / 48204) 3 AS expected_if_independent;</pre>		
1 row		
<table><thead><tr><th>expected_if_independent</th></tr></thead><tbody><tr><td>0.04091713370059525</td></tr></tbody></table>	expected_if_independent	0.04091713370059525
expected_if_independent		
0.04091713370059525		

$$P(\text{Congestion}) \times P(\text{Clear Weather}) = 0.040917 \approx 4.09\%$$

$$P(\text{Congestion} \cap \text{Clear Weather}) = 1,763 / 48,204 = 0.036574 \approx 3.66\%$$

Independence conclusion

For independence, these would need to be equal:

$$0.0366 \neq 0.0409$$

Therefore:

$$P(\text{High Temp} | \text{Congestion}) = \frac{1,867}{7,100} = \mathbf{0.2630}$$

Clear weather and congestion **cannot be considered independent** in this dataset. If the two events were independent, the probability of both occurring together would be approximately 4.09%. The actual joint probability is lower, at approximately 3.66%, indicating an association between clear weather conditions and congestion.

```

1 SELECT
2     weather_main,
3     COUNT(*) AS total,
4     SUM(traffic_volume > 5500) AS congested
5 FROM traffic
6 WHERE weather_main IN ('Clear', 'Clouds')
7 GROUP BY weather_main;

```

2 rows

weather_main	total	congested
Clear	13391	1763
Clouds	15164	2592

```

1 SELECT
2     ROUND((1763.0 / 11628) / (2592.0 / 12572), 3)
3     AS odds_ratio;

```

1 row

The odds of congestion during clear weather are approximately 0.735 times the odds during cloudy weather, or about 26.5% lower.

The probability analysis suggests that weather has some association with congestion, although the relationship isn't strong enough to establish causation. Congestion occurred in 14.73% of all observations, while 24.83% of congested observations occurred during clear weather. The actual probability of congestion and clear weather occurring together (3.66%) differs from the expected probability under independence (4.09%), suggesting that clear weather and congestion aren't independent in this dataset. In addition, the odds ratio of 0.735 indicates that the odds of congestion were approximately 26.5% lower in clear weather than in cloudy weather. However, traffic conditions are also influenced by factors such as time of day, day of week, holidays and seasonal patterns, so these results shouldn't be interpreted as evidence that cloudy weather causes congestion.